



Separation of polyethylene glycols and amino-terminated polyethylene glycols by high-performance liquid chromatography under near critical conditions

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ARTICLE INFO

Article history:

Received 3 February 2016

Received in revised form 11 April 2016

Accepted 13 April 2016

Available online 14 April 2016

Keywords:

Poly(ethylene glycol)

Amino-substituted PEG

Liquid chromatography at critical conditions

Reversed-phase liquid chromatography
Ion-exchange

ABSTRACT

The separation and characterization of polyethylene glycols (PEGs) and amino-substituted derivatives on common silica-based reversed-phase packing columns using isocratic elution is described. This separation is achieved by liquid chromatography under the near critical conditions (LCCC), based on the number of amino functional end groups without obvious effect of molar mass for PEGs. The mobile phase is acetonitrile in water with an optimal ammonium acetate buffer. The separation mechanism of PEG and amino-substituted PEG under the near LCCC on silica-based packing columns is confirmed to be ion-exchange interaction. Under the LCCC of PEG backbone, with fine tune of buffer concentration, the retention factor ratios for benzylamine and phenol in buffered mobile phases, $\alpha(\text{benzylamine/phenol})$ -values, were used to assess the ion-exchange capacity on silica-based reversed-phase packing columns. To the best of our knowledge, this is the first report on separation of amino-functional PEGs independent of the molar mass by isocratic elution using common C18 or phenyl reversed-phase packing columns.

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1. Introduction

There has been a great interest in polyethylene glycol (PEG) and their conjugates of biologically active molecules in recent years [1–5], because PEG is one of the few polymers approved for clinical internal applications in humans by FDA (US Food and Drug Administration) [6]. The first step in preparation of functional PEGs is the chemical derivatization of end groups. Amino-substituted PEG derivatives are widely used among the various functionalized PEGs [7–9]. As the non-functionalized portion is an impurity that will not be reactive towards the target PEGylation, the fractions of non-, mono- and bi-functional PEGs need to be determined for raw materials in the production of polymer conjugates for pharmaceutical applications. However, there are limited numbers of publications on separation and characterization of PEGs and functional PEGs [10]. For example, separation of modified PEG methyl ether on a cyanopropyl column was reported with a gradient elution, with low detection sensitivity and poor shapes of peaks [11]. Barman

et al. [12] described very good reversed-phase high-performance liquid chromatography (RP-HPLC) method for separation and quantitative determination of PEG impurities in two monofunctional PEG types, PEG methyl ether and PEG vinyl ether. Separation of PEGs and amino-substituted PEGs on a TSK-GEL G4000PW_{XL} column was described, but molar mass effect of PEG backbone was not mentioned [13]. Tang et al. [14] demonstrated the separation and detection of bi-maleimide-PEG and mono-maleimide-PEG by RP-HPLC gradient elution, although molar mass effect of PEG backbone was observed clearly.

LCCC (liquid chromatography at critical conditions) has been proven to be especially effective in the analysis of functional polymers. In this mode, the retention volume of non-functional polymer becomes independent of molar mass, and this offers the opportunity to separate polymers with respect to their functionality [15–22]. However, the exact LCCC is not easy to obtain experimentally, and the LCCC conditions do not necessarily provide good separation of polymers with different end-groups [20]. RP-HPLC is by far the most widely used mode of HPLC, in which, stationary phases based on modified silica are most frequently used [23,24]. This kind of modification is seldom completed, the residual silanol groups may affect separation, especially in the analysis of basic compounds [25,26]. Cox and Stout had investigated the retention mechanisms of basic compounds on silica and C₈ reversed-phase packing, which were shown to be mainly ion exchange mechanism

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Table 1

Structures and abbreviations of the studied PEGs ($\text{RCH}_2\text{CH}_2\text{O}[\text{CH}_2\text{CH}_2\text{O}]_n\text{CH}_2\text{CH}_2\text{R}'$).

Compound	R	R'	Avg. molar mass	Abbreviations
1	OH	OH	5500–7000	PEG 6k
2	NH ₂	NH ₂	6120 (M_w), 5828 (M_n)	PEGDNH ₂ 6k
3	CH ₃ O	OH	5272 (M_w), 4677 (M_n)	MPEGOH 5k
4	CH ₃ O	NH ₂	4957 (M_w), 4812 (M_n)	MPEGNH ₂ 5k
5	OH	OH	3500–4500	PEG 4K
6	OH	NH ₂	4120 (M_w), 4079 (M_n)	PEGNH ₂ 4k
7	NH ₂	NH ₂	4312 (M_w), 4106 (M_n)	PEGDNH ₂ 4k
8	OH	OH	1900–2200	PEG 2k
9	NH ₂	NH ₂	2090 (M_w), 2029 (M_n)	PEGDNH ₂ 2k

M_w : weight-average molar mass, M_n : number-average molar mass.

[24,27,28]. Euerby et al. [29] had evaluated a range of commercially available reversed-phase liquid chromatographic columns containing phenyl moieties and C18 moieties, using ion-exchange capacity (IEC) measured as the retention ratio $\alpha_{\text{A/P}}$, $k_{\text{benzylamine}}/k_{\text{phenol}}$, to assess the active amount of ion exchange sites on silica-based surface in the specific buffered mobile phase. The greater the values of $\alpha_{\text{A/P}}$, the larger amount of ion exchange sites, and the smaller the values, the less amount of ion-exchange sites [30–32].

To the best of our knowledge, no detailed studies on separation of amino-functional PEGs independent of the molar mass by isocratic LCCC using reversed-phase packing columns have been published yet. In this work, we describe a simple and suitable LC method for separation of PEGs and amino-functional PEGs independent of the molar mass by isocratic elution, taking advantage of ion exchange sites of common silica-based reversed-phase packing columns at appropriate LCCC conditions. Specially, the optimization of the buffer composition in the mobile phase will be investigated.

2. Materials and methods

2.1. Materials

Characteristics and abbreviations of the polymer samples used in this work are summarized in Table 1. PEG 2k, PEG 4k and PEG 6k were purchased from Sinopharm Chemical Reagent (Shanghai, China). MPEGOH 5k was purchased from Sigma-Aldrich (Steinheim, Germany). PEGDNH₂ 2k, PEGNH₂ 4k, PEGDNH₂ 4k, MPEGNH₂ 5k and PEGDNH₂ 6k were from Beijing Chemgen Pharma (Beijing, China). Ammonium acetate from Sinopharm Chemical Reagent (AR grade, Shanghai, China) was used. Acetonitrile was purchased from Tedia Company (HPLC grade, Fairfield, USA). Water from an Arrium® pro UF purification system (Sartorius, Goettingen, Germany) was used. All chemicals and reagents were used as received.

2.2. Chromatographic separation and detection

Analytical chromatography was performed by an HPLC system assembled with Cometro 6000 LDI pump (South Plainfield, USA), a temperature controller and a Softa (Model 300 S, Burbank, USA) evaporative light scattering detector (ELSD) on a XB-Phenyl column (250 × 4.6 mm I.D., 5 μm particle size, 300 Å pore size, Welch Materials, Shanghai, China), a Shodex-C18 column (250 × 4.6 mm I.D., 5 μm particle size, 100 Å pore size, Showa Denko, Tokyo, Japan), or a TSK-GEL G4000PWXL column (300 × 7.8 mm I.D., 10 μm particle size, 500 Å pore size, Tosoh, Tokyo, Japan). PEGs and amino-substituted PEG derivatives do not possess any structural elements absorbing in the commonly applied UV region. ELSD is especially well-suited for the determination of any nonvolatile analyte [33]. For ELSD, air was used as carrier gas, the temperatures of the drift tube and spray chamber were set at 80 °C and 50 °C, respectively.

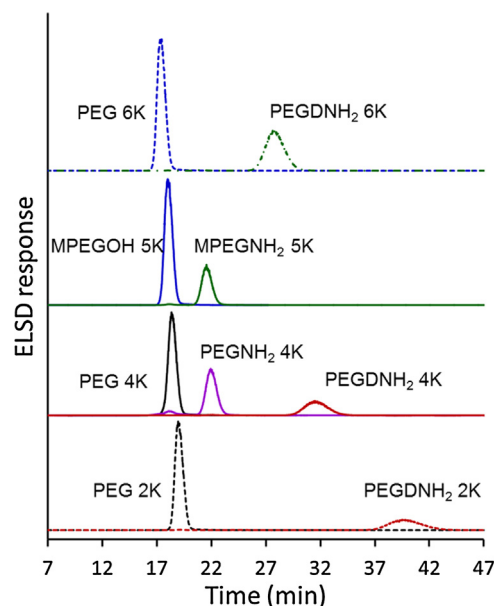


Fig. 1. The chromatograms of PEGs and amino-substituted PEG derivatives obtained on TSK-GEL G4000PWXL column (mobile phase: water with 2.5 mmol/L $\text{CH}_3\text{COONH}_4$, flow-rate 0.5 mL/min, polymer concentration in water: 2 mg/mL).

The use of a volatile buffer salt was obligatory because of the ELSD, therefore $\text{CH}_3\text{COONH}_4$ was chosen as the buffer salt in this study.

The mobile phases were prepared by first dissolving appropriate amounts of $\text{CH}_3\text{COONH}_4$ in water for the desired concentration. The aqueous phase was filtered through a 0.22 μm HPLC filter before mixed with the organic phase and degassed prior to use by ultrasound. Samples were prepared in a 35:65 (volume ratio) mixture of acetonitrile and water at the concentration of 0.2 mg/mL (unless mentioned otherwise) and injected by a loop volume of 20 μL. Acetone was used as a t_0 marker. Data was collected using EZChrom Elite software. Data analysis was performed on a Microsoft EXCEL spreadsheet. Principal component analysis (PCA) was applied to classify the polymers according to the retention times measured for PEGs and amino-substituted PEG derivatives by MATLAB 2011a.

3. Results and discussion

Good separation of PEGs and the amino-substituted PEG derivatives with an average molar mass of 2000 and 3350 according to the ion-exchange interaction between amino-terminated group and the column packing surface was reported [13] on a TSK-GEL G4000PWXL column, which is widely used in aqueous size-exclusion chromatography (SEC). However, the molar mass effect of PEGs was not mentioned. Actually, the size exclusion effect for PEG backbone was clearly observed, just as shown in Fig. 1. It can be seen that sample PEG 6k with higher molar mass was eluted earlier than samples PEG 2k and PEG 4k with lower molar mass. Bi-amino PEGDNH₂ 6k also had a shorter retention time compared to other bi-amino-substituted PEGs (PEGDNH₂ 4k and PEGDNH₂ 2k). Thus separation methods of PEGs and amino-substituted PEG derivatives avoiding the effect of molar mass are worthy to explore.

3.1. Effects of organic solvent composition on the retention of PEG

Liquid chromatography at critical conditions (LCCC) has been proven to be especially effective in the analysis of functional polymers. In this mode, the retention volume of non-functional polymer becomes independent of molar mass, and this offers the opportunity to separate polymers with respect to their functionality

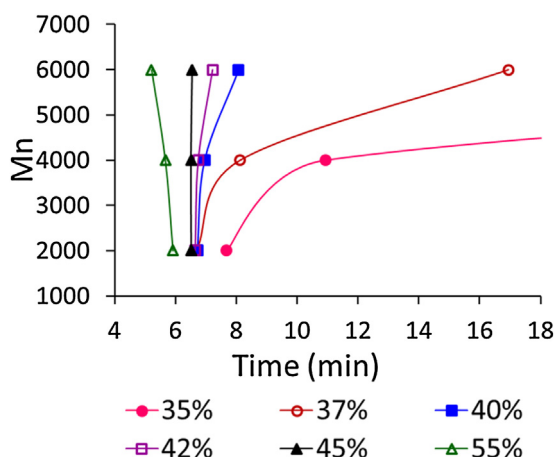


Fig. 2. The retention time of PEGs with different molar masses on XB-Phenyl column in various compositions of acetonitrile-water at 30 °C (The values indicated in the figure refer to the vol% of acetonitrile in water).

[34,35]. To obtain the LCCC, the effect of the composition of mobile phase (volume percentage of acetonitrile in water) on the interaction between PEG chain and XB-Phenyl column packing was investigated and shown in Fig. 2. Critical composition for PEG was found at 45% acetonitrile in water at 30 °C. At this critical composition, nearly identical elution time for the lower molar mass PEG 2k and the higher molar mass PEG 6k was obtained, which indicates that the molar mass does not strongly influence the retention at the selected critical chromatographic condition. This could provide the opportunity to separate amino-substituted PEG derivatives according to the number of amino-functional end-groups. The LCCC for PEG on another silica-based reversed-phase packing column Shodex-C18 column was obtained with the mobile phase 40% vol acetonitrile in water at 25 °C.

3.2. The role of the ion exchange properties in silica-based stationary phases

XB-Phenyl column and Shodex-C18 column are both silica-based reversed-phase packing columns. This kind of packing possesses ion exchange properties which can play a role in separation of amines via electrostatic interaction through the ion exchange sites on the surface of the packing [25]. Ion-exchange capacity (IEC) could be measured as retention factor ratio $\alpha_{A/P}$ ($k_{\text{benzylamine}}/k_{\text{phenol}}$). The $\alpha_{A/P}$ value can reflect the active amount of ion exchange sites on silica-based surface in the specific buffered mobile phase. The greater the values of $\alpha_{A/P}$, the larger amount of ion exchange sites, and the lower $\alpha_{A/P}$ -values imply the absence of ion-exchange sites [30].

Different $\alpha_{A/P}$ values were obtained for different $\text{CH}_3\text{COONH}_4$ concentrations in the critical mobile phases of PEGs on both silica-based columns and are summarized in Table 2. It can be seen from this Table that the $\alpha_{A/P}$ -values decreased with increasing buffer $\text{CH}_3\text{COONH}_4$ contents. Stella et al. [36] had pointed out that if

Table 2

The values of $\alpha_{A/P}$ obtained on XB-Phenyl column and Shodex-C18 column.

$[\text{CH}_3\text{COONH}_4]$ (mmol/L)	$\alpha_{A/P}$ (XB-Phenyl ^a)	$\alpha_{A/P}$ (Shodex-C18 ^b)
1	$\alpha_{A/P} = 2.18$	$\alpha_{A/P} = 0.47$
2	$\alpha_{A/P} = 1.04$	$\alpha_{A/P} = 0.19$
5	$\alpha_{A/P} = 0.29$	$\alpha_{A/P} = -0.09$
6	$\alpha_{A/P} = 0.19$	–
10	$\alpha_{A/P} = 0.03$	$\alpha_{A/P} = -0.13$

^a Mobile-phase containing 45% acetonitrile 30 °C.

^b Mobile-phase containing 40% acetonitrile 25 °C.

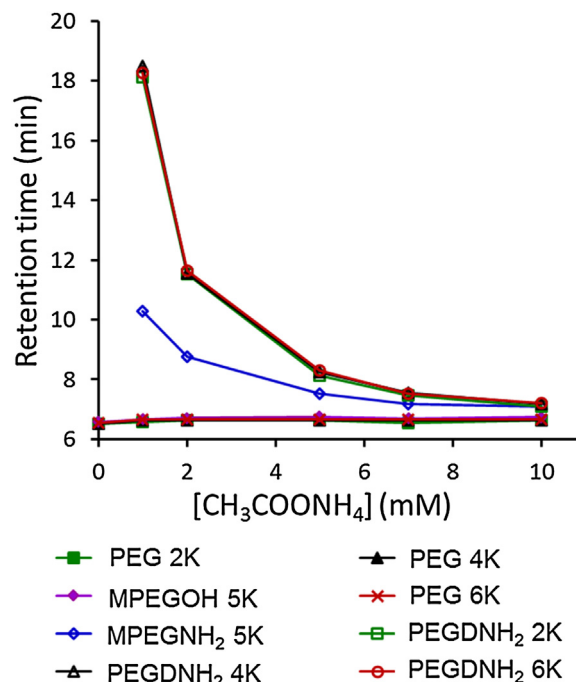


Fig. 3. Effect of ammonium acetate concentration in the mobile phase (45% acetonitrile in water) on retention time of PEGs on XB-phenyl column.

the values of $\alpha_{A/P}$ were larger than 0.1, then ion exchange sites on the surface of the silica-based column were active. Based on data in Table 2, if the $\text{CH}_3\text{COONH}_4$ concentration is lower than or equal to 6 mmol/L on XB-Phenyl column (2 mmol/L for Shodex-C18 column), these ion exchange sites offered electrostatic interaction with the amino functional group, which provided the opportunity to separate amino-substituted PEG derivatives under near LCCC.

3.3. Effects of the buffer concentration on the separation of amino-substituted PEG derivatives

The silica-based columns used in the study can provide ion exchange interaction with amino group according to the $\alpha_{A/P}$ -values. Amino-substituted PEGs were used to study the interaction between amino functional group and ion exchange sites under near LCCC of PEG. Different $\text{CH}_3\text{COONH}_4$ concentrations were used on XB-phenyl column to study the effects of salt on separation of various PEGs, when mobile phase contained 45% acetonitrile. As shown in Fig. 3, the retention times for all non-amino-substituted PEGs with different molar mass were almost same when the mobile phases containing $\text{CH}_3\text{COONH}_4$ from 1 to 10 mmol/L were used. The retention time of mono-amino-substituted MPEGNH₂ 5k gradually increased with the decrease of salt concentrations. The retention times of bi-amino-substituted PEGs with different molar mass also increased with reducing buffer concentrations. It is worth noting that the bi-amino-substituted PEGs had more dramatic changes in retention time than mono-amino-substituted PEGs, when the concentrations of $\text{CH}_3\text{COONH}_4$ varied in the mobile phases.

According to the ion-exchange mechanism [27], a plot of retention factor (k') against the inverse of counter-ion concentration should be a straight-line graph. Such plots of k' against the inverse of ammonium ion concentration for mono-amino-substituted MPEGNH₂ and bi-amino-substituted PEGs are shown in Fig. 4. Indeed, all of these plots are linear, manifesting that the LC interaction mechanism is ion-exchange under the near LCCC with 45% acetonitrile in water. Bi-amino-substituted PEGs with different molar masses had higher slopes than mono-amino-substituted

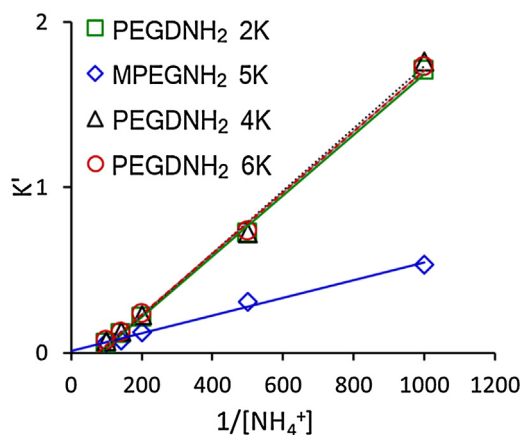


Fig. 4. Plot of retention factor (k') against inverse ammonium ion concentration on XB-phenyl column.

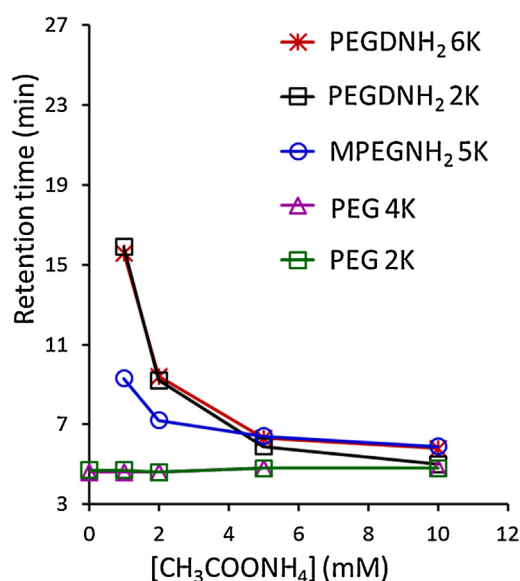


Fig. 5. Effect of ammonium acetate concentration in the mobile phase (40% acetonitrile in water) on retention time of PEGs on Shodex-C18 column.

MPEGNH₂ 5k, which indicated that bi-amino-substituted PEG had stronger ionic interaction with the column surface and longer retention time.

The effects of different CH₃COONH₄ concentrations on separation of PEGs was studied with Shodex-C18 column under the near LCCC conditions for PEG, containing 40% acetonitrile in water at 25 °C. As shown in Fig. 5, similar separation results for amino-substituted PEG derivatives based on the number of amino-functional end-groups were obtained.

3.4. Chemometric evaluation of retention data

Principal component analysis (PCA) is a frequently used statistical method. It can extract information from large datasets to reveal similarities and differences among the columns and test substances [37,38]. Engelhardt et al. [39] used PCA to classify the differentiation between the amines that studied. The PCA of PEGs and amino-substituted PEG derivatives was shown in Fig. 6, with the ammonium acetate concentration in the mobile phase from 0 to 10 mmol/L. It is obvious that the ammonium acetate concentration used in the mobile phase had a stronger influence on the retention times of amino-substituted PEGs than those of

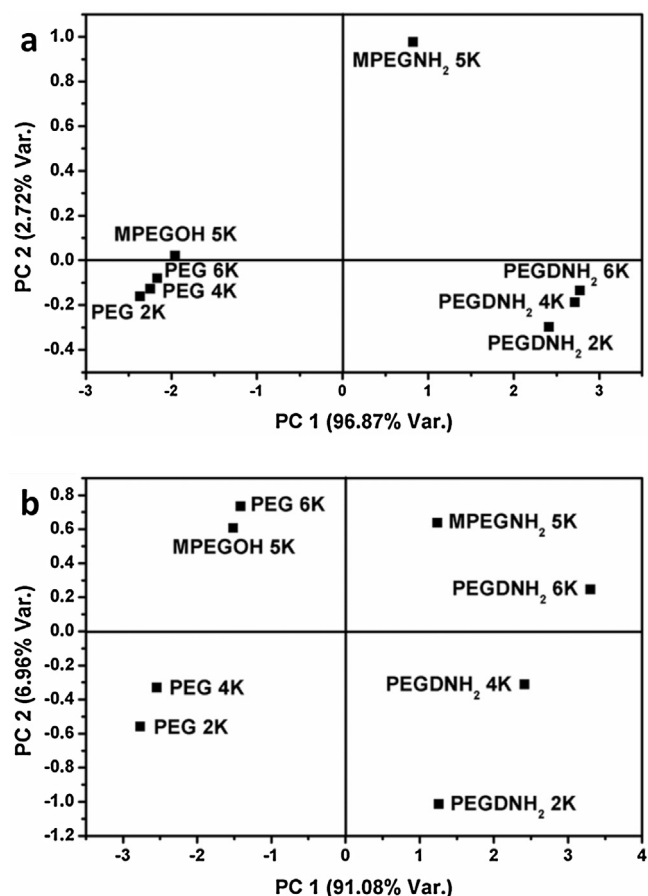


Fig. 6. Principal component analysis: distribution of PEGs and amino-substituted PEG derivatives with different ammonium acetate concentration in the mobile phase (a) acetonitrile/water (45/55, V/V) and (b) acetonitrile/water (42/58, V/V). Column: XB-Phenyl at 30 °C.

non-amino-substituted PEGs. Specially, distribution of PEGs and amino-substituted PEG derivatives under LCCC (Fig. 6a) clearly depends on the type of functional end-groups without obvious molar mass effect. In Fig. 7 a and b with ammonium acetate concentration of 2 and 5 mmol/L, respectively, the grouping of PEGs and amino-substituted PEG derivatives can be clearly seen. However, in Fig. 7 c and d with ammonium acetate concentration of 7 and 10 mmol/L, respectively, no differentiation between PEGs and amino-substituted PEG derivatives is possible, which means that PEGs and amino-substituted PEG derivatives cannot be separated according to the end groups under these chromatographic conditions.

3.5. Chromatograms of amino-substituted PEG derivatives

The chromatograms of PEGs on XB-Phenyl column are shown in Fig. 8 as the representative adsorption mode for PEG backbone, when the mobile phase contained 42% acetonitrile in water with 2 mmol/L CH₃COONH₄ at 30 °C. It was clearly shown that PEGs were separated mainly according to the number of amino-functional end-groups. However, the molar mass effect on retention time is obviously observed. For example, PEG 2k eluted faster than PEG 6k, PEGDNH₂ 2k came out earlier than PEGDNH₂ 6k. MPEGOH 5k had the longest retention time in non-amino-substituted PEG samples investigated, probably due to the hydrophobic effect of methyl group [12].

To avoid the molar mass effect in separation of PEGs, the LCCC was used. The chromatograms of PEGs are shown in Fig. 9

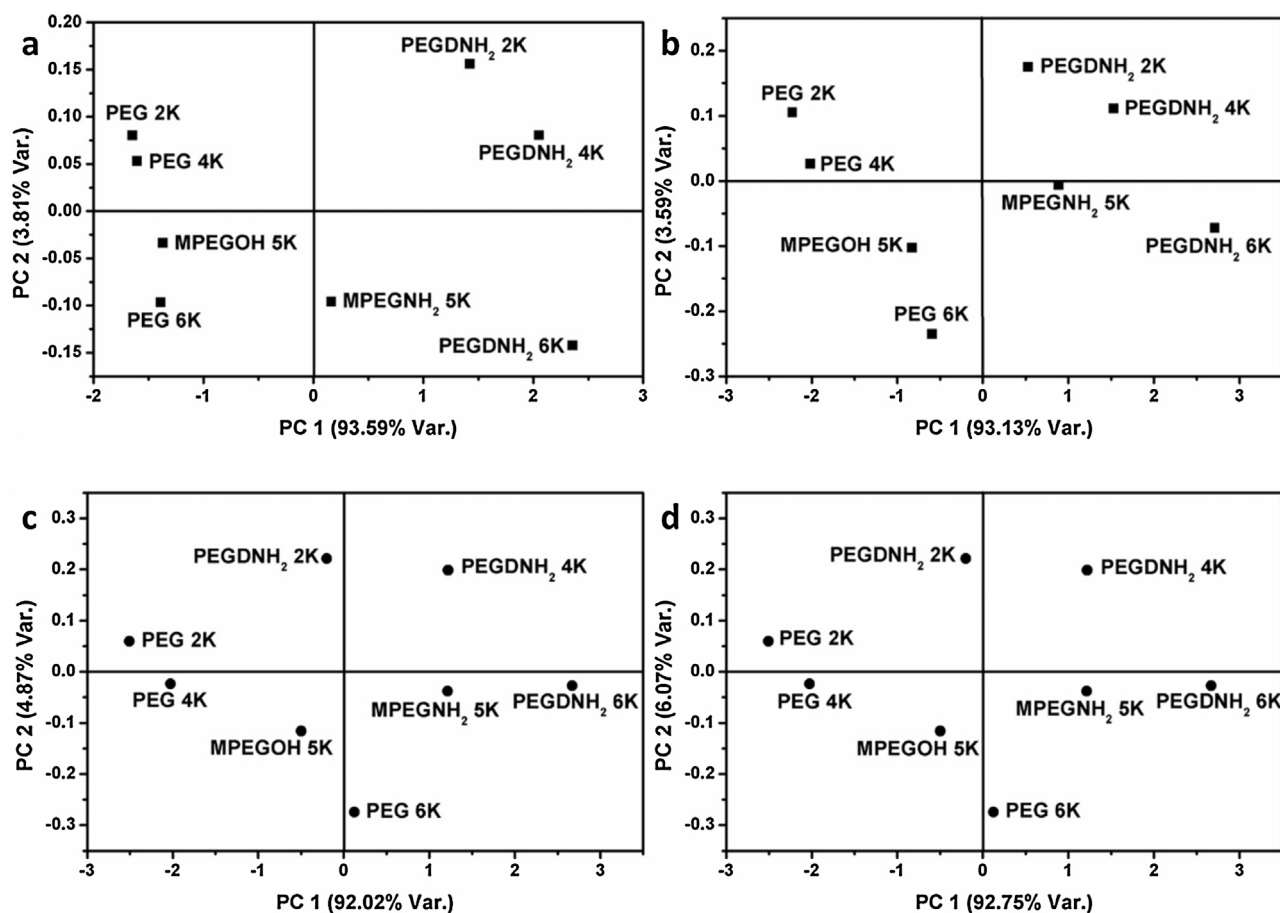


Fig. 7. Principal component analysis: distribution of PEGs and amino-substituted PEG derivatives with different column temperature at 25°C, 30°C and 35°C in the acetonitrile/water (42/58, V/V) mobile phase with (a) 2, (b) 5, (c) 7 and (d) 10 mmol/L ammonium acetate. Column: XB-Phenyl.

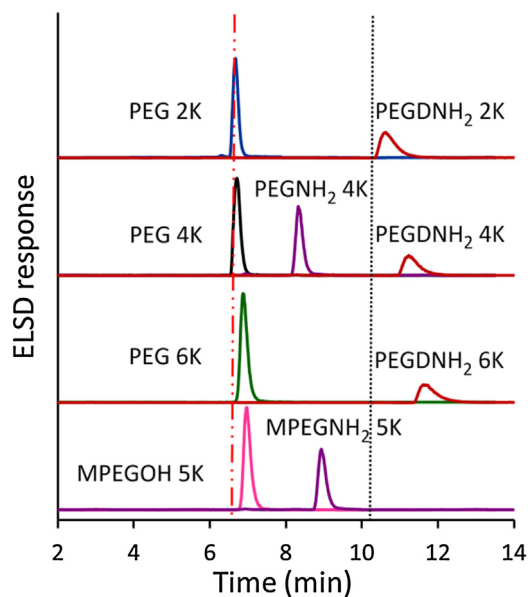


Fig. 8. The chromatograms of PEGs obtained on XB-Phenyl column (mobile phase: acetonitrile: water = 42:58 with 2 mmol/L $\text{CH}_3\text{COONH}_4$, flow-rate 0.5 mL/min).

when mobile phase contained 45% acetonitrile in water with 2 mmol/L $\text{CH}_3\text{COONH}_4$ at 30°C. It was clearly shown that PEGs were successfully separated according to the number of amino functional groups. Their molar mass had no obvious impact on retention

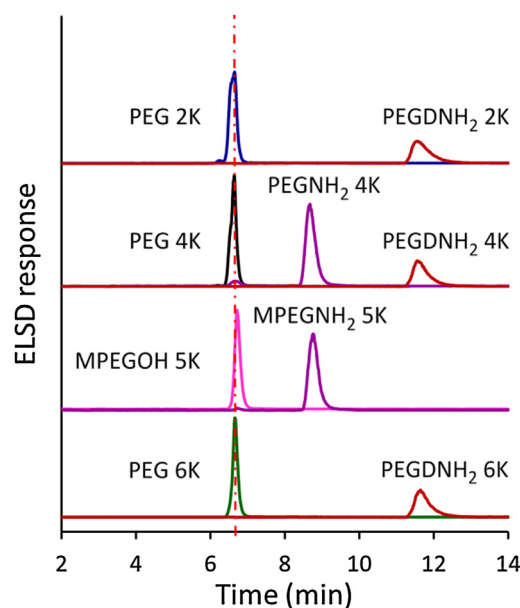


Fig. 9. The chromatograms of PEGs obtained on XB-Phenyl column (mobile phase: acetonitrile: water = 45:55 with 2 mmol/L $\text{CH}_3\text{COONH}_4$, flow-rate 0.5 mL/min).

time. For example, the retention times were almost same for PEG 2k, PEG 4k, MPEGOH 5k and PEG 6k, and the retention times for PEGDNH₂ 2k, PEGDNH₂ 4k and PEGDNH₂ 6k were also similar. But mono-amino-substituted PEGNH₂ 4k and MPEGNH₂ 5k had longer

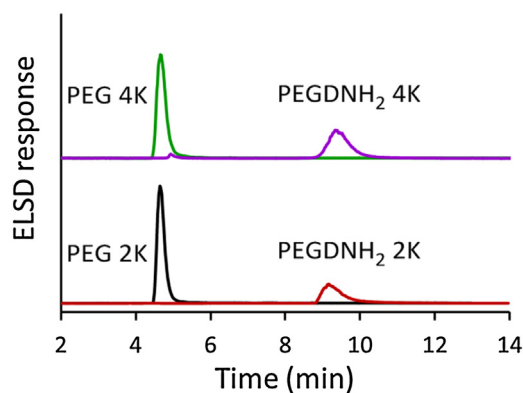


Fig. 10. The chromatograms of PEGs obtained on Shodex-C18 column (mobile phase: acetonitrile: water = 40:60 with 2 mmol/L $\text{CH}_3\text{COONH}_4$, flow-rate 0.5 mL/min).

retention time than non-amino-substituted PEG but less retention time than bi-amino-substituted PEG, because amino functional end-group could interact with the column silanol ion exchange sites. The interaction became stronger when the number of amino-group increased, so PEGDNH₂ 2k, PEGDNH₂ 4k and PEGDNH₂ 6k had the longest retention time among these PEG samples. This chromatographic condition worked well in separation of PEGs based on the number of amino functional groups independent of the molar mass on XB-phenyl column. The chromatograms of PEGs on another silica-based Shodex-C18 column are shown in Fig. 10. It can be seen that PEGs were successfully separated according to the number of amino functional group without obvious effect of molar mass.

4. Conclusions

Critical conditions of liquid chromatography for PEGs with different molar mass on silica-based reversed-phase columns were first obtained by adjusting mobile phase composition with various percentage of organic solvent acetonitrile in water. Under these LCCC, with fine tune of buffer concentration, the retention factor ratios for benzylamine and phenol in buffered mobile phases, $\alpha_{A/P}$ values, were used to assess the ion-exchange capacity on silica-based reversed-phase packing column. It was shown that the separation was mainly due to ion-exchange mechanism through ion-exchange sites on silica-based packing columns. After determination of the suitable buffer concentration (2 mmol/L $\text{CH}_3\text{COONH}_4$, volatile buffer) under the near LCCC, it was possible to separate the PEGs according to the number of amino functional end group without obvious effect of molar mass. PCA was used to classify the differentiation between the polymers that studied. The distribution of PEGs and amino-substituted PEG derivatives was observed clearly, depending on the type of functional groups under LCCC. So far as we know, this is the first report on separation of amino-functional PEGs independent of the molar mass by isocratic elution using common reversed-phase packing columns.

Acknowledgements

This research was financially supported by the National Natural Science Foundation of China (21374083 and 21174109). We thank Mr. Jian Yang (State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University) for his cooperation in principal component analysis and Miss Yibo

Jiang (Wageningen University) for her English editing of the revised version.

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